Name ____ Date ___________ Period _____

Formulating Absolute Value Linear Equations

Algebra II | Unit 2 | Day 4 | TEK A2.6.D

Objective: I can formulate (set up, without solving) an absolute value linear equation from a real-world description.

Vocabulary

Formulate — set up an equation from a description - ________________________.

Center value — the target/ideal number the situation is measured against.

Margin — how far off from the center is still acceptable.

Template — every problem in this lesson fits the pattern $|x - \text{center}| = \text{margin}$.

Part 1 — Clarissa's New House

Clarissa wants a house within 5 houses of house #1248 on an even-numbered street.

Center value: ___________ Margin: ______

Equation: ________________________

Part 2 — Surveying Margin of Error

A surveying instrument measures a 10-mile stretch with a 1.5% margin of error.

The 1.5% margin of error is measured as a percent of the true 10-mile distance x, so it is written $1.5 = |x - 10|/10 \cdot 100$.

Equation: ___________________________________

Part 3 — Brighton Builders' Hallway

A hallway has offices on both sides, evenly spaced from the center at 15x feet, and the total layout must equal 292 feet with 52 feet reserved for walls and doors.

Equation: ____________________________

Practice — Formulate, Do Not Solve

1. A recipe calls for 2 cups of flour, with acceptable measurement error of 0.25 cups.

Center: ______ Margin: ________ Equation: ________________________

2. A shipping company's box-weight target is 50 lb, with a tolerance of 3 lb.

Equation: ______________________

3. A factory's bolt length must be within 0.02 cm of 6 cm.

Equation: ________________________

Exit Ticket

A thermostat targets 70°F, with an acceptable range of 2°F above or below. Formulate an equation - do not solve.

Center: ________ Margin: ______

Equation: ______________________